

Using (1) and (2) to solve for $u_b = 1.998 \text{ m s}^{-1}$

6 B Take moments about the pivot.

$$\text{Torque} = 5.0(2)(0.5 \sin 40^\circ) = 3.2 \text{ N m}$$

7 A Consider the vertical component of the tension and letting the angle that the cord makes with the metal sheet